Mean Value Theorems & Series Expansions

1. Verify Cauchy's mean value theorem for the following functions:
   $f(x) = \sqrt{x}, g(x) = \frac{1}{\sqrt{x}}$ $in\ [a,b], where\ 0 < a < b.$
2. Verify Rolle's theorem for $f(x) = x(x-1)(x-2)$ in $[0,2]$.
3. Verify Lagrange's Mean Value Theorem (LMVT) for the function
   $f(x) = (x-1)(x-2)(x-3)$ in $(0,4)$.
4. If $f(x)$ and $g(x)$ are respectively e^x and e^{-x}, prove that 'c' of Cauchy's mean value theorem is the arithmetic mean between a and b.
5. Verify Rolle's theorem for $f(x) = log\left[\frac{x^2+ab}{(a+b)x}\right]$ in $[a,b]$ where $b > a > 0$.
6. If $f(x) = log\ x$, $g(x) = x^2$ in $[a,b]$ with $b > a > 1$, using Cauchy's mean value theorem prove that: $\frac{log\ b - log\ a}{b-a} = \frac{a+b}{2c^2}$
7. Verify LMVT for $f(x) = x^3 - x^2 - 5x + 3$ in $[0,4]$.
8. Verify Rolle's theorem for $f(x) = |x|$ on $[-1,1]$.
9. Verify Cauchy's mean value theorem for $f(x) = x^2$ and $g(x) = x^3$ on $[1,2]$.
10. Using Mean Value Theorem show that: $\frac{\pi}{4} + \frac{3}{25} < tan^{-1}\left(\frac{4}{3}\right) < \frac{\pi}{4} + \frac{1}{6}$
11. If x is positive, show that: $\frac{x}{1+x} < log(1+x) < x$
12. Using LMVT, find the value of $\sqrt[5]{245}$.
13. Expand $tan^{-1}x$ about origin.
14. Find the Maclaurin's series expansion of $f(x) = sin^2x$ and hence find the approximate value of $sin^2 16^o$.
15. By using Mean Value Theorem, prove that: $|sin\ u - sin\ v| \leq |u - v|$
16. Expand $tan^{-1}x$ in powers of $(x-1)$ up to 4th degree term.
17. Find Taylor's series expansion of $f(x) = cos\ x$ about $x = \frac{\pi}{6}$ and hence find the value of $cos\ 32^o$.
18. Find the Maclaurin's expansion of $log(1 + e^x)$.